

Remark. It is worthwhile to point out that the only places where the hypothesis of simple-connectivity entered in the proof were in the uses of the logarithm and the square root. Thus it would have sufficed to have assumed (in addition to the hypothesis that Ω is proper) that Ω is **holomorphically simply connected** in the sense that for any holomorphic function f in Ω and any closed curve γ in Ω , we have $\int_{\gamma} f(z) dz = 0$. Further discussion of this point, and various equivalent properties of simple-connectivity, are given in Appendix B.

4 Conformal mappings onto polygons

The Riemann mapping theorem guarantees the existence of a conformal map from any proper, simply connected open set to the disc, or equivalently to the upper half-plane, but this theorem gives little insight as to the exact form of this map. In Section 1 we gave various explicit formulas in the case of regions that have symmetries, but it is of course unreasonable to ask for an explicit formula in the general case. There is, however, another class of open sets for which there are nice formulas, namely the polygons. Our aim in this last section is to give a proof of the Schwarz-Christoffel formula, which describes the nature of conformal maps from the disc (or upper half-plane) to polygons.

4.1 Some examples

We begin by studying some motivating examples. The first two correspond to easy (but infinite and degenerate) cases.

EXAMPLE 1. First, we investigate the conformal map from the upper half-plane to the sector $\{z : 0 < \arg z < \alpha\pi\}$, with $0 < \alpha < 2$, given in Section 1 by $f(z) = z^{\alpha}$. Anticipating the Schwarz-Christoffel formula below, we write

$$z^{\alpha} = f(z) = \int_0^z f'(\zeta) d\zeta = \alpha \int_0^z \zeta^{-\beta} d\zeta$$

with $\alpha + \beta = 1$, and where the integral is taken along any path in the upper half-plane. In fact, by continuity and Cauchy's theorem, we may take the path of integration to lie in the closure of the upper half-plane. Although the behavior of f follows immediately from the original definition, we study it in terms of the integral expression above, since this provides insight for the general case treated later.

Note first that $\zeta^{-\beta}$ is integrable near 0 since $\beta < 1$, therefore $f(0) = 0$. Observe that when z is real and positive ($z = x$), then $f'(x) = \alpha x^{\alpha-1}$ is

positive; also it is *not* finitely integrable at ∞ . Therefore, as x travels from 0 to ∞ , we see that $f(x)$ increases from 0 to ∞ , thus f maps $[0, \infty)$ to $[0, \infty)$. On the other hand, when $z = x$ is negative, then

$$f'(z) = \alpha|x|^{\alpha-1}e^{i\pi(\alpha-1)} = -\alpha|x|^{\alpha-1}e^{i\pi\alpha},$$

so f maps the segment $(-\infty, 0]$ to $(e^{i\pi\alpha}\infty, 0]$. The situation is illustrated in Figure 3 where the infinite segment A is mapped to A' and the segment B is mapped to B' , with the direction of travel indicated in Figure 3.



Figure 3. The conformal map z^α

EXAMPLE 2. Next, we consider for $z \in \mathbb{H}$,

$$f(z) = \int_0^z \frac{d\zeta}{(1-\zeta^2)^{1/2}},$$

where the integral is taken from 0 to z along any path in the closed upper half-plane. We choose the branch for $(1-\zeta^2)^{1/2}$ that makes it holomorphic in the upper half-plane and positive when $-1 < \zeta < 1$. As a result

$$(1-\zeta^2)^{-1/2} = i(\zeta^2-1)^{-1/2} \quad \text{when } \zeta > 1.$$

We observe that f maps the real line to the boundary of the half-strip pictured in Figure 4.

In fact, since $f(\pm 1) = \pm\pi/2$, and $f'(x) > 0$ if $-1 < x < 1$, we see that f maps the segment B to B' . Moreover

$$f(x) = \frac{\pi}{2} + \int_1^x f'(x) dx \quad \text{when } x > 1, \quad \text{and} \quad \int_1^\infty \frac{dx}{(x^2-1)^{1/2}} = \infty.$$

Thus, as x travels along the segment C , the image traverses the infinite segment C' . Similarly segment A is mapped to A' .

Note the connection of this example with Example 8 in Section 1.2. In fact, one can show that the function $f(z)$ is the inverse to the function

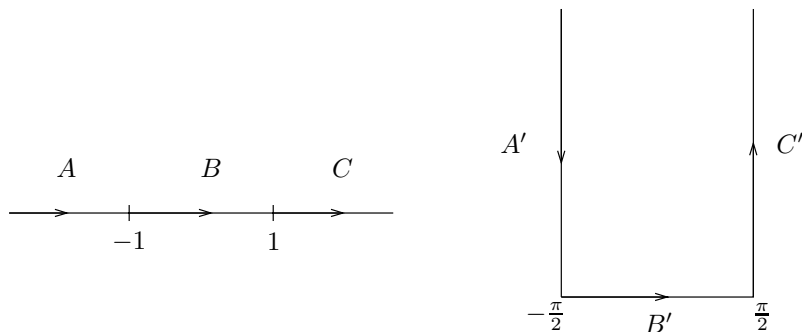


Figure 4. Mapping of the boundary in Example 2

$\sin z$, and hence f takes $\mathbb{H}$ conformally to the interior of the half-strip bounded by the segments A' , B' , and C' .

EXAMPLE 3. Here we take

$$f(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}, \quad z \in \mathbb{H},$$

where k is a fixed real number with $0 < k < 1$ (the branch of $[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}$ in the upper half-plane is chosen to be the one that is positive when ζ is real and $-1 < \zeta < 1$). Integrals of this kind are called **elliptic integrals**, because variants of these arise in the calculation of the arc-length of an ellipse. We shall observe that f maps the real axis onto the rectangle shown in Figure 5(b), where K and K' are determined by

$$K = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \quad K' = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}}.$$

We divide the real axis into four “segments,” with division points $-1/k$, -1 , 1 , and $1/k$ (see Figure 5(a)). The segments are $[-1/k, -1]$, $[-1, 1]$, $[1, 1/k]$, and $[1/k, -1/k]$, the last consisting of the union of the two half-segments $[1/k, \infty)$ and $(-\infty, -1/k]$. It is clear from the definitions that $f(\pm 1) = \pm K$, and since $f'(x) > 0$, when $-1 < x < 1$, it follows that f maps the segment $[-1, 1]$ to $[-K, K]$. Moreover, since

$$f(z) = K + \int_1^x \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}} \quad \text{if } 1 < x < 1/k,$$

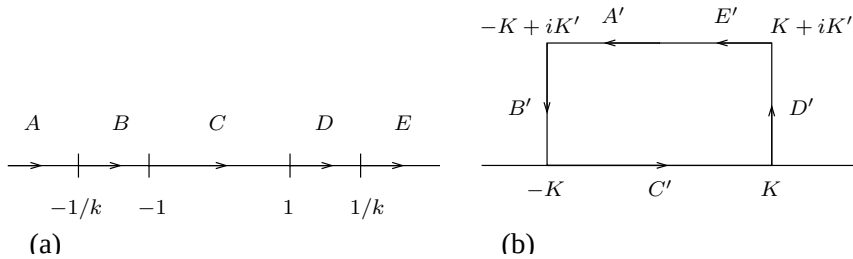


Figure 5. Mapping of the boundary in Example 3

we see that f maps the segment $[1, 1/k]$ to $[K, K + iK']$, where K' was defined above. Similarly, f maps $[-1/k, -1]$ to $[-K + iK', -K]$. Next, when $x > 1/k$ we have

$$f'(x) = -\frac{1}{[(x^2 - 1)(k^2x^2 - 1)]^{1/2}},$$

and therefore,

$$f(x) = K + iK' - \int_{1/k}^x \frac{dx}{[(x^2 - 1)(k^2x^2 - 1)]^{1/2}}.$$

However,

$$\int_{1/k}^{\infty} \frac{dx}{[(x^2 - 1)(k^2x^2 - 1)]^{1/2}} = \int_0^1 \frac{dx}{[(1 - x^2)(1 - k^2x^2)]^{1/2}},$$

as can be seen by making the change of variables $x = 1/ku$ in the integral on the left. Thus f maps the segment $[1/k, \infty)$ to the segment $[K + iK', iK')$. Similarly f maps $(-\infty, -1/k]$ to $[-K + iK', iK')$. Altogether, then, f maps the real axis to the above rectangle, with the point at infinity corresponding to the mid-point of the upper side of the rectangle.

The results obtained so far lead naturally to two problems.

The first, which we pursue next, consists of a generalization of the above examples. More precisely we define the Schwarz-Christoffel integral and prove that it maps the real line to a polygonal line.

Second, we note that in the examples above little was inferred about the behavior of f in $\mathbb{H}$ itself. In particular, we have not shown that f maps $\mathbb{H}$ conformally to the interior of the corresponding polygon. After a careful study of the boundary behavior of conformal maps, we

prove a theorem that guarantees that the conformal map from the upper half-plane to a simply connected region bounded by a polygonal line is essentially given by a Schwarz-Christoffel integral.

4.2 The Schwarz-Christoffel integral

With the examples of the previous section in mind, we define the general **Schwarz-Christoffel integral** by

$$(5) \quad S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}.$$

Here $A_1 < A_2 < \cdots < A_n$ are n distinct points on the real axis arranged in increasing order. The exponents β_k will be assumed to satisfy the conditions $\beta_k < 1$ for each k and $1 < \sum_{k=1}^n \beta_k$.⁶

The integrand in (5) is defined as follows: $(z - A_k)^{\beta_k}$ is that branch (defined in the complex plane slit along the infinite ray $\{A_k + iy : y \leq 0\}$) which is positive when $z = x$ is real and $x > A_k$. As a result

$$(z - A_k)^{\beta_k} = \begin{cases} (x - A_k)^{\beta_k} & \text{if } x \text{ is real and } x > A_k, \\ |x - A_k|^{\beta_k} e^{i\pi\beta_k} & \text{if } x \text{ is real and } x < A_k. \end{cases}$$

The complex plane slit along the union of the rays $\cup_{k=1}^n \{A_k + iy : y \leq 0\}$ is simply connected (see Exercise 19), so the integral that defines $S(z)$ is holomorphic in this open set. Since the requirement $\beta_k < 1$ implies that the singularities $(\zeta - A_k)^{-\beta_k}$ are integrable near A_k , the function S is continuous up to the real line, including the points A_k , with $k = 1, \dots, n$. Finally, this continuity condition implies that the integral can be taken along any path in the complex plane that avoids the union of the *open* slits $\cup_{k=1}^n \{A_k + iy : y < 0\}$.

Now

$$\left| \prod_{k=1}^n (\zeta - A_k)^{-\beta_k} \right| \leq c |\zeta|^{-\sum \beta_k}$$

for $|\zeta|$ large, so the assumption $\sum \beta_k > 1$ guarantees the convergence of the integral (5) at infinity. This fact and Cauchy's theorem imply that $\lim_{r \rightarrow \infty} S(re^{i\theta})$ exists and is independent of the angle θ , $0 \leq \theta \leq \pi$. We call this limit a_∞ , and we let $a_k = S(A_k)$ for $k = 1, \dots, n$.

⁶Note that the case $\sum \beta_k \leq 1$, which occurs in Examples 1 and 2 above is excluded. However, a modification of the proposition that follows can be made to take these cases into account; but then $S(z)$ is no longer bounded in the upper half-plane.